

Unwrapping the mind: Knowledge, Cognition and Artificial Intelligence

Confessions of a Computational Linguist: What We Lost and Gained in the Age of Large Language Models



Simone Ponzetto

- Chair of Information Systems III
- NLP and IR group @ UniMa
- Data and Web science group



Research interest

- Artificial Intelligence
- Language and computation

... and their application in

- (Digital) Humanities
- (Computational) Social Science

Background: linguistics => AI => NLP

Data and Web Science Group

A large group focused on Data and Web Science:

- **Web-based Systems** (C. Bizer)
- **Data Analytics** (R. Gemulla)
- **Web Data Mining** (H. Paulheim)
- **Natural Language Processing** (me)
- **Artificial Intelligence** (H. Stuckenschmidt)
- **Computer Vision** (M. Keuper)
- <https://dws.informatik.uni-mannheim.de>



Cognitive Science Initiative (CSI): Mannheim



... you should know these people by now!

Today's menu

- What is Natural Language Processing?
- The Linguistic Legacy
- The Data Revolution: When Scale Trumped Structure
- The Confession: Performance Without Linguistics?

Learning outcomes

By the end of this lecture, you should know:

- at least roughly what is Natural Language Processing :)
- a few examples of research on automated text processing that we do in Mannheim in our NLP group
- different approaches developed throughout the history of (automated) Machine Translation
- Different ways to represent language within these approaches

This is meant the basis for an open discussion in the end

List of reading materials (relevant for the exam)

IMPORTANT: Please use this slide deck and read these papers for the exam

- Richard Sutton: The Bitter Lesson

<http://www.incompleteideas.net/IncIdeas/BitterLesson.html>

- Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the Dangers of Stochastic Parrots: Can Language Models Be Too Big? . In Proceedings of FAccT '21.

<https://dl.acm.org/doi/pdf/10.1145/3442188.3445922>

WHAT IS NATURAL LANGUAGE PROCESSING?

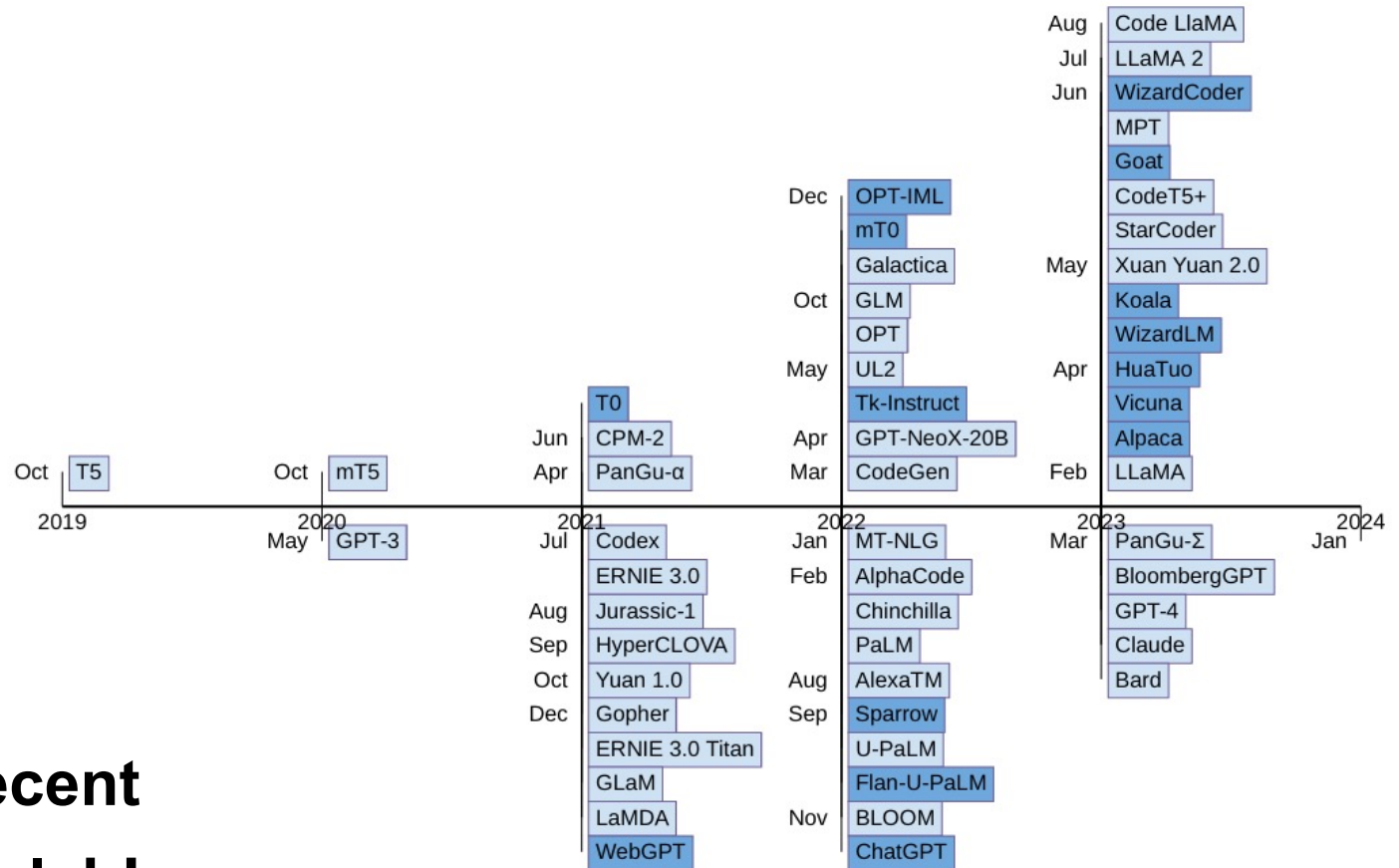
What is Natural Language Processing?

- I needed such an introduction slide for the past 20 years...

- Not anymore!

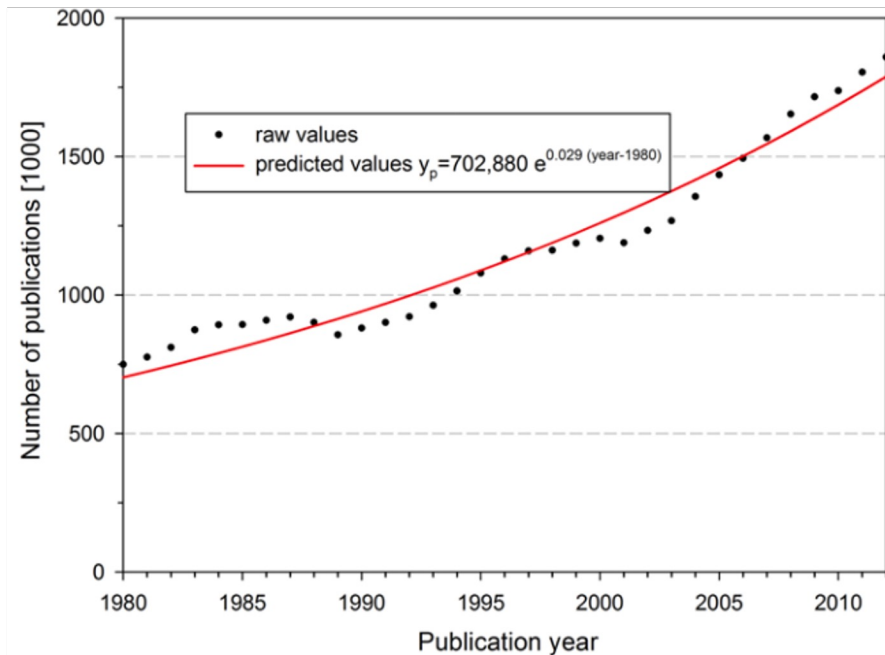
- Enter LLMs

- **Next stop: some recent highlights from our lab!**



Research topic: SciNLP

The number of publication is rapidly increasing [Bornmann and Mutz 2015]



Bornmann and Mutz (2015): Growth rates of modern science:
A bibliometric analysis based on the number of publications and cited references.

Some applications we might want to enable...

- Citation Recommendation
- Scholarly Document Summarization

Scholarly Document “Extreme” Summarization

- Create a short summary of the research paper, (i.e., TLDR: “too long; didn’t read”) that covers its main points
- X-SCITLDR (Takeshita et al., JCDL 2022)

Abstract: We propose a method for meta-learning reinforcement learning algorithms by searching over the space of computational graphs which compute the loss function for a value based model-free RL agent to optimize. [...]

Introduction: Designing new deep reinforcement learning algorithms that can efficiently solve across a wide variety of problems generally requires a tremendous amount of manual effort. [...]

Conclusion: In this work, we have presented a method for learning reinforcement learning algorithms. We design a general language for representing algorithms which compute the loss function for [...]

TLDR: We meta-learn RL algorithms by evolving computational graphs which compute the loss function for a value-based model free RL agent to optimize.

ACLSum (Takeshita et al., NAACL 2024)

- Aspect-based summarization dataset with extractive and abstractive annotations
- Annotated and validated by domain experts to achieve high quality
- LIVE DEMO: <https://gengo.sotaro.io>

Extractive Summary

Statistical Machine Translation with a Factorized Grammar

Libin Shen and Bing Zhang and Spyros Matsoukas and
Jiwei Xu and Ralph Weischedel
Raytheon BBN Technologies
Cambridge, MA 02138, USA
{lshen, bzhang, smatsouk, jxu, weisschedel}@bbn.com

Abstract

In modern machine translation practice, a statistical phrasal or hierarchical translation system usually relies on a large set of translation rules extracted from bilingual training data. This approach not only results in space and efficiency issues, but also suffers from the sparse data problem. In this paper, we propose to use factorized grammars as an alternative to the traditional phrasal or hierarchical translation rules. We designed a method to take advantage of the XTAG English Grammar to facilitate the extraction of factorized rules. We report on various steps of low-resource language translation, and showed consistent significant improvement in BLEU over state-of-the-art using dependency-based systems with 200K words of bilingual training data.

1 Introduction

A statistical phrasal (Koehn et al., 2003; Och and Ney, 2004) or hierarchical (Chiang, 2006; Maru et al., 2006) machine translation system usually relies on a very large set of translation rules extracted from bilingual training data with heuristic methods on word alignment results. According to our own experience, we obtain about 200K of rules from training data of about 50M words on each side. This immediately becomes an engineering challenge on space and search efficiency.

A common practice to circumvent this problem is to filter the rules based on development sets in the step of rule extraction or before the decoding phase, instead of building a real distributed system. However, this strategy only works for research systems,

for which the segments for translation are always fixed.

However, do we really need such a large rule set to represent information from the training data of much smaller size? Linguists in the grammar construction field already showed us a perfect solution to a similar problem. The answer is to use a factorized grammar. Linguists decompose lexicalized linguistic structures into two parts, (lexicalized) templates and lexical items. Templates are further organized into families. Each family is associated with a set of lexical items which can be used to lexicalize all the templates in this family. For example, the XTAG English Grammar (XTAG-Group, 2001), a hand-crafted grammar based on the Tree Adjoining Grammar (TAG) (Joshi and Schabes, 1997) formalism, is a grammar of this kind, which employs factorization with LTAG c-tree templates and lexical items.

Factorized grammars not only relieve the burden on space and search, but also alleviate the sparse data problem, especially for low-resource language translation with few training data. With a factored model, we do not need to observe exact “template – lexical item” occurrences in training. New rules can be generated from template families and lexical items either offline or on the fly, explicitly or implicitly. In fact, the factorization approach has been successfully applied on the morphological level in previous study on MT (Koehn and Hoang, 2007). In this work, we will go further to investigate factorized

transformation rules by exploiting the rich XTAG English Grammar.

We evaluate the effect of using factorized translation grammars on various steps of low-resource language translation, since low-resource MT suffers greatly on poor generalization capability of trans-

616

Proceedings of the 2010 Conference on Empirical Methods in Natural Language Processing, pages 616–625, MIT, Massachusetts, USA, 9–11 October 2010. ©2010 Association for Computational Linguistics

Abstractive Summary

Challenge

Existing statistical phrasal or hierarchical machine translation system relies on a large set of translation rules which results in engineering challenges.

Approach

The proposed method consistently outperforms existing methods in BLEU on various low-resource language translation tasks with less training data.

Outcome

They propose to use factorized grammar from the field of linguistics as more general translation rules from XTAG English Grammar.

GenGO! (Takeshita et al., 2024)

[Omakase](#)[Read Later](#)

 Search...



Welcome to *GenGO*!

Make sure to checkout our new prototype system, **GenGO-Chat**, a LLM-powered RAG system on top of GenGO database for NLPers.

ACL Events

AACL				2022		2020		
ACL	2025	2024	2023	2022	2021	2020	2019	2018
COLING	2025			2022		2020		2018
EMNLP		2024	2023	2022	2021	2020	2019	2018
Findings	2025	2024	2023	2022	2021	2020		
LREC		2024		2022		2020		2018
*SEM		2024	2023	2022	2021	2020		
TACL	2025		2023	2022	2021	2020		
EACL		2024	2023		2021			
NAACL	2025	2024		2022	2021		2019	2018

An example of how a paper looks like in *GenGO*.

GenGO: ACL Paper Explorer with Semantic Features (2024.acl-demos)   

Sotaro Takeshita, Simone Ponzetto, Kai Eckert

Challenge: Scholarly document processing (SDP) is a powerful tool for researchers to process knowledge stored in research papers.

Approach: They propose a system that allows researchers to search papers published in ACL conferences with metadata and text embeddings.

Outcome: The proposed system is simple and efficient to reduce maintenance and financial costs and is extensible to support open development and transparency.

GenGO! (Takeshita et al., 2024)

GenGO »

The AI Gap: How Socioeconomic Status Affects Language Technology Interactions

(2025.acl-long)

bassignana-etal-2025-ai

Elisa Bassignana, Amanda Cercas Curry, Dirk Hovy

SES

DIGITAL TECHNOLOGIES

LARGE LANGUAGE MODELS

LLMS

GENERATIVE LANGUAGE TECHNOLOGIES

PROXY METRICS

LANGUAGE TECHNOLOGY USAGE

Challenge:

Socioeconomic status (SES) fundamentally influences how people interact with technology, but it is limited by proxy metrics and synthetic data.

Approach:

They collect 6,482 prompts from previous interactions of 1,000 individuals from 'diverse socioeconomic backgrounds' about their use of language technologies and generative AI.

Outcome:

The findings show that higher SES groups have higher levels of abstraction, convey requests more concisely, and topics like 'inclusivity' and 'travel'.

Abstract

▼

Similar Papers

Overview

Challenge

Approach

Outcome

Classist Tools: Social Class Correlates with Performance in NLP

(2024.acl-long)

Amanda Curry, Giuseppe Attanasio, Zeerak Talat, Dirk Hovy

Text Error Correction

Challenge:

despite growing concerns surrounding fairness and bias in NLP, there is a dearth of studies delving into the effects it may have on NLP systems.

Approach:

They argue that NLP systems' performance is affected by speakers' SES, potentially disadvantaging less-privileged socioeconomic groups.

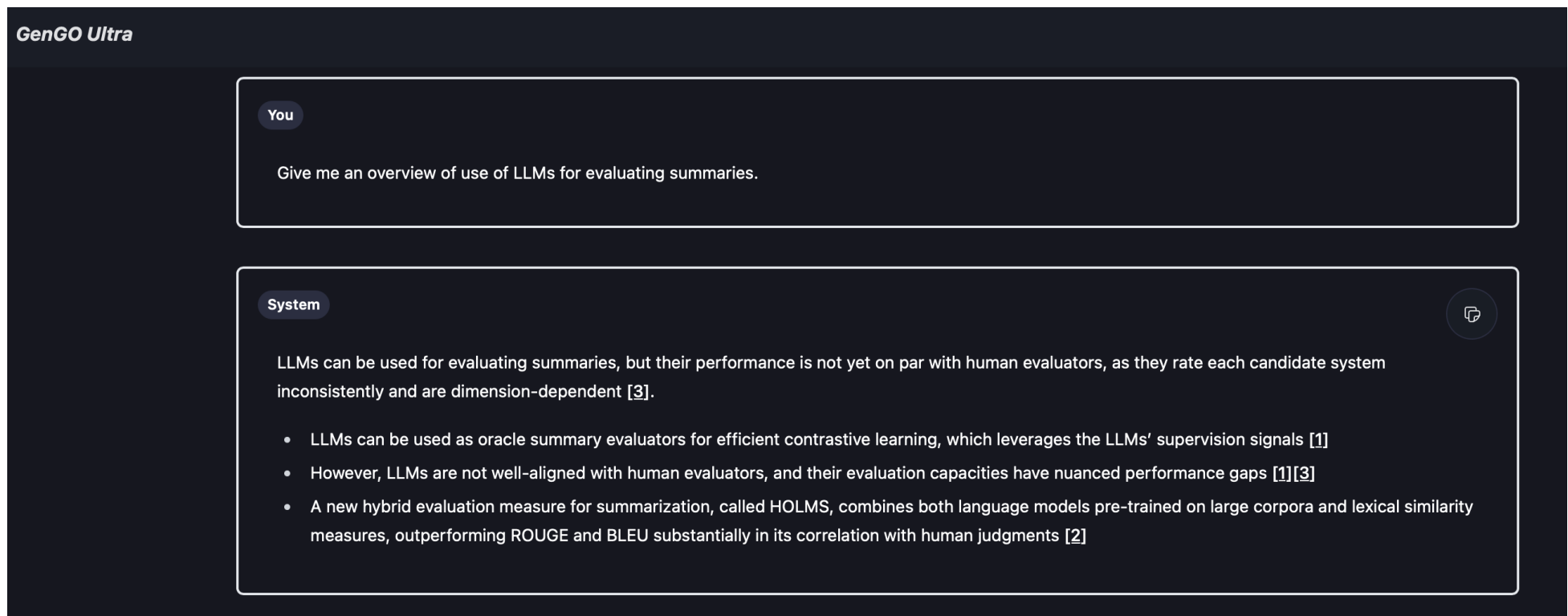
Outcome:

The proposed model shows that NLP systems perform better on tasks with social class, ethnicity and geographical variation than those without social class.

Universität Mannheim – Unwrapping the mind (Ponzetto) – Slide 14

GenGO Ultra (Takeshita et al., 2025)

- RAG: retrieve and generate answer from relevant external sources (like documents or databases)
- DEMO: <https://gengo-ultra.sotaro.io>



The screenshot displays the GenGO Ultra web interface. At the top left, the text "GenGO Ultra" is visible. The interface features a dark background with white text. A user input box, labeled "You", contains the text "Give me an overview of use of LLMs for evaluating summaries." Below this, a system response box, labeled "System", contains a paragraph and a bulleted list. The paragraph states: "LLMs can be used for evaluating summaries, but their performance is not yet on par with human evaluators, as they rate each candidate system inconsistently and are dimension-dependent [3].". The bulleted list includes three items: "LLMs can be used as oracle summary evaluators for efficient contrastive learning, which leverages the LLMs' supervision signals [1]", "However, LLMs are not well-aligned with human evaluators, and their evaluation capacities have nuanced performance gaps [1][3]", and "A new hybrid evaluation measure for summarization, called HOLMS, combines both language models pre-trained on large corpora and lexical similarity measures, outperforming ROUGE and BLEU substantially in its correlation with human judgments [2]". A copy icon is located in the top right corner of the system response box.

GenGO Ultra

You

Give me an overview of use of LLMs for evaluating summaries.

System

LLMs can be used for evaluating summaries, but their performance is not yet on par with human evaluators, as they rate each candidate system inconsistently and are dimension-dependent [3].

- LLMs can be used as oracle summary evaluators for efficient contrastive learning, which leverages the LLMs' supervision signals [1]
- However, LLMs are not well-aligned with human evaluators, and their evaluation capacities have nuanced performance gaps [1][3]
- A new hybrid evaluation measure for summarization, called HOLMS, combines both language models pre-trained on large corpora and lexical similarity measures, outperforming ROUGE and BLEU substantially in its correlation with human judgments [2]

Not only *natural* language: multimodal LLMs

- Research question: **how to leverage multimodal LLMs for scientific figures?**



Observation 1

Even though **sketching** ideas on paper is **easy**, creating **high-quality** scientific figures can be **time-consuming** and **challenging**.

Observation 2

Recreating existing figures that are stored in low-level formats (JPG, PNG, PDF, SVG, ...) which do not preserve semantic information is **equally complex**.

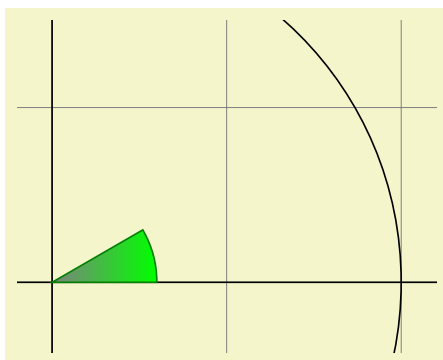
From sketches to graphics programs

We introduce **DETikZIFY**, a multimodal LLM that automatically synthesizes scientific figures based on sketches and existing figures as semantics-preserving **TikZ graphics programs**. We leverage TikZ because it is:

expressive can represent complex figures with few commands

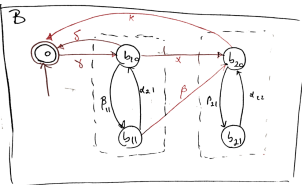
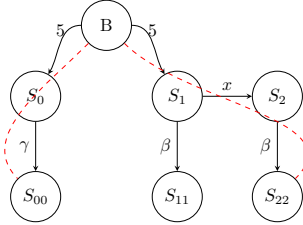
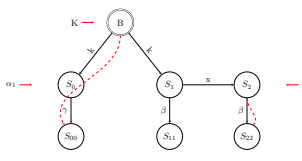
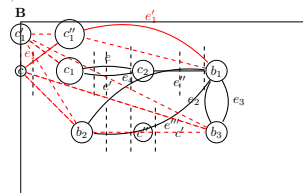
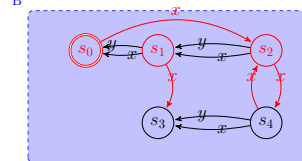
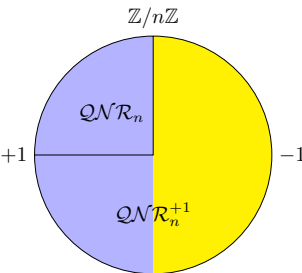
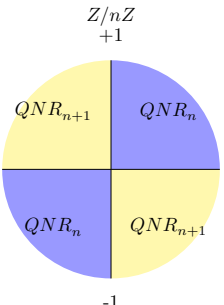
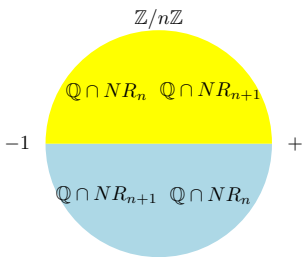
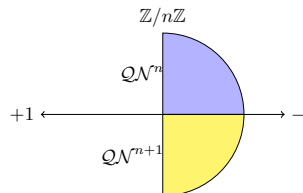
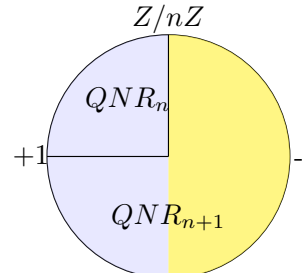
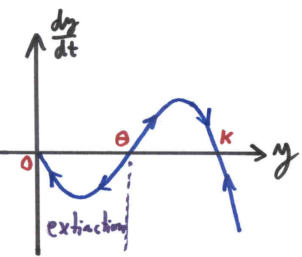
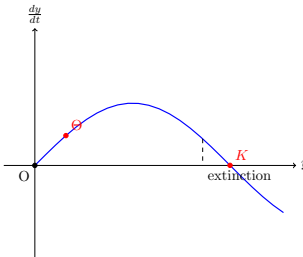
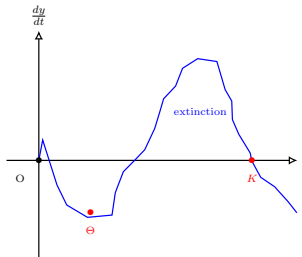
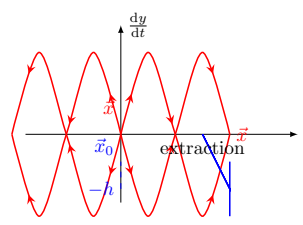
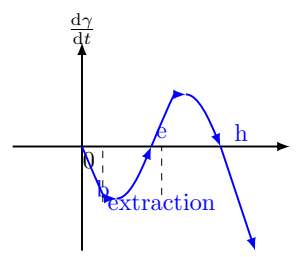
versatile supports a wide range of figure types (and more)

popular most frequently discussed topic on T_EX.SE



```
\begin{tikzpicture}[scale=3]
  \clip (-0.1,-0.2) rectangle (1.1,0.75);
  \draw[step=.5cm,gray,very thin] (-1.4,-1.4) grid (1.4,1.4);
  \draw (-1.5,0) -- (1.5,0);
  \draw (0,-1.5) -- (0,1.5);
  \draw (0,0) circle (1cm);
  \shadedraw[left color=gray,right color=green, draw=green!50!black]
    (0,0) -- (3mm,0mm) arc (0:30:3mm) -- cycle;
\end{tikzpicture}
```

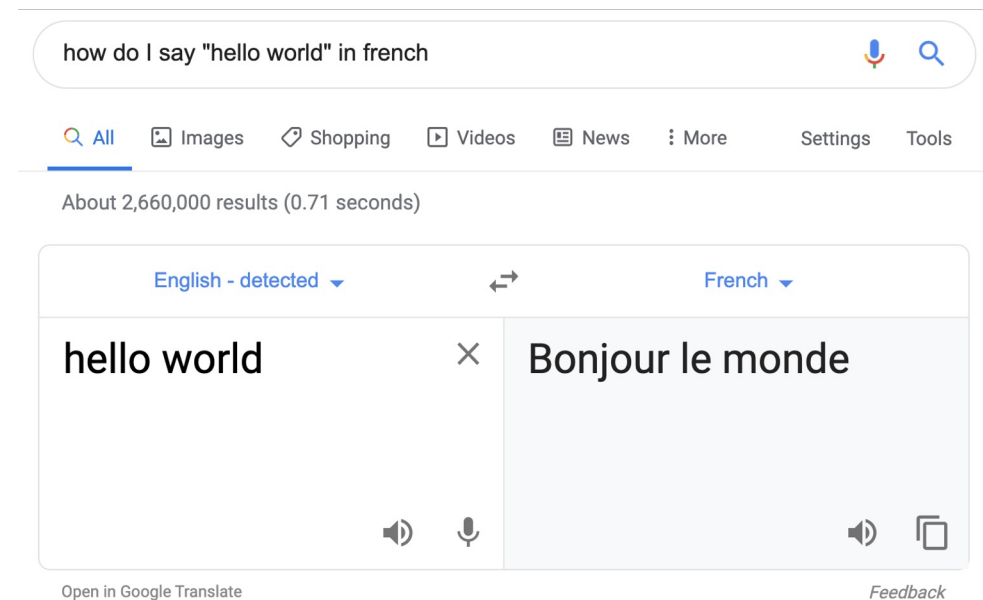
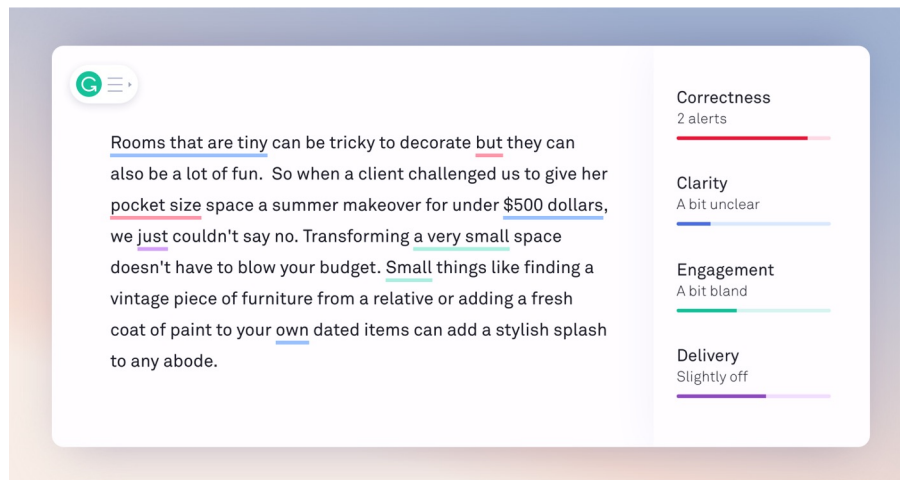
DeTikZify: Examples

Input	GPT-4V	GPT-4V (IR)	DETIKZIFY	DETIKZIFY (IR)
				
				
				

– DEMO: <https://github.com/potamides/DeTikZify>

Let's zoom out: a definition of Natural Language Processing

- Methods to **automatically process (i.e., understand and generate) natural language data**



NLP Tasks

- NLP is a broad area that studies a wide range of tasks.
- Some examples of NLP tasks:
 - Question answering (QA)
 - Machine translation (MT)
 - Text summarization
 - Assisted writing
- Useful to distinguish them by their inputs and outputs

Question Answering (QA)

×  

Bilder Videos News Bücher Maps Flüge Finanzen

Ungefähr 39.600.000 Ergebnisse (0,65 Sekunden)

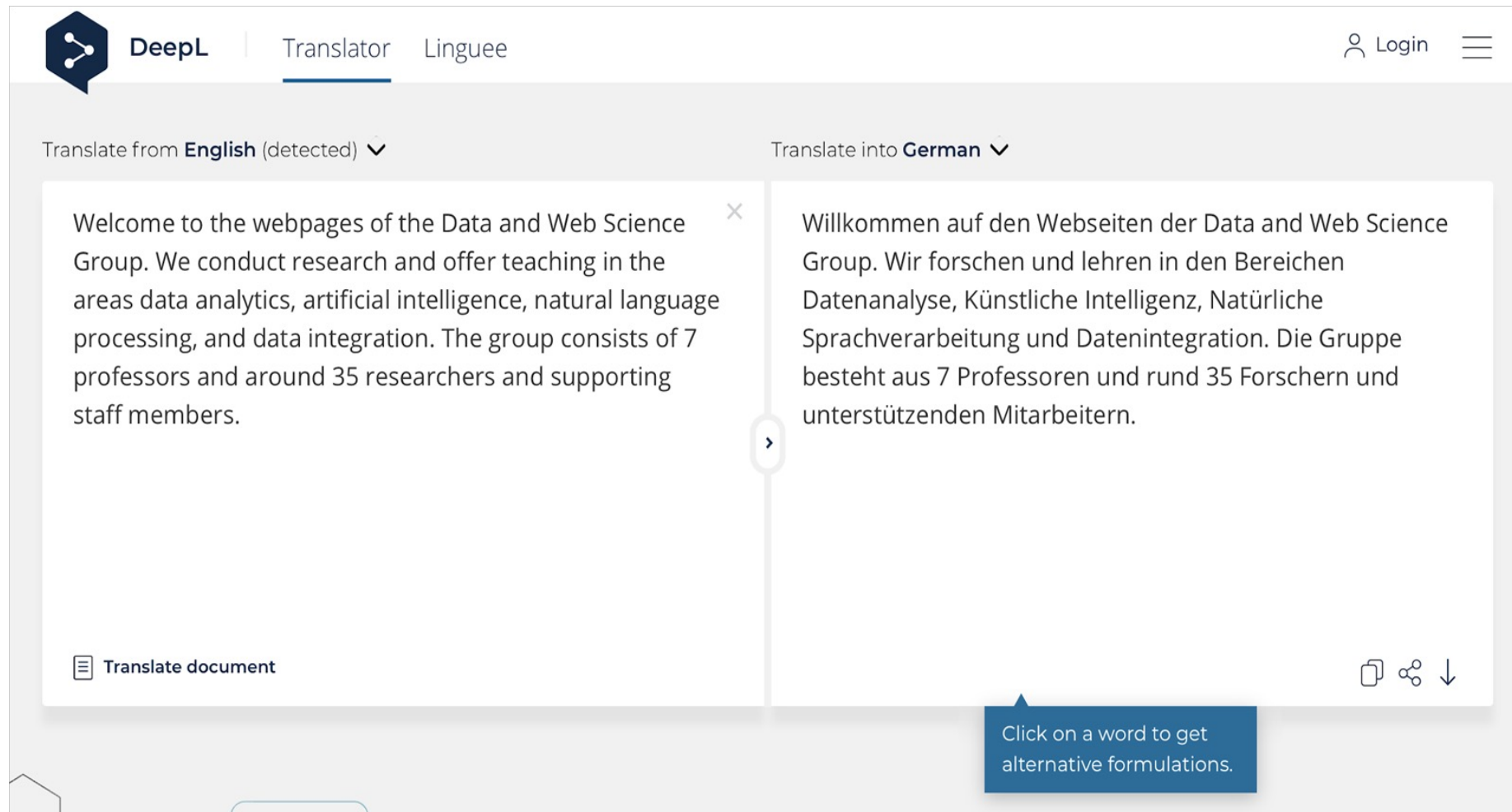
William Shakespeare / Geburtsdatum

April 1564



William Shakespeare ['wɪljəm 'ʃeɪkspɪə] (getauft am 26. April 1564^{jul.} in Stratford-upon-Avon; gestorben am 23. April^{jul.} / 3. Mai 1616^{greg.} ebenda) war ein englischer Dichter, Theaterunternehmer und Schauspieler, dessen Dramen zu den bedeutendsten Werken der Weltliteratur gehören.

Machine Translation (MT)



The screenshot displays the DeepL Translator web interface. At the top, the DeepL logo is on the left, and 'Translator' and 'Linguee' are in the center. On the right, there is a 'Login' button and a menu icon. Below the header, the interface is split into two main sections: 'Translate from English (detected)' on the left and 'Translate into German' on the right. The English text on the left reads: 'Welcome to the webpages of the Data and Web Science Group. We conduct research and offer teaching in the areas data analytics, artificial intelligence, natural language processing, and data integration. The group consists of 7 professors and around 35 researchers and supporting staff members.' The German translation on the right reads: 'Willkommen auf den Webseiten der Data and Web Science Group. Wir forschen und lehren in den Bereichen Datenanalyse, Künstliche Intelligenz, Natürliche Sprachverarbeitung und Datenintegration. Die Gruppe besteht aus 7 Professoren und rund 35 Forschern und unterstützenden Mitarbeitern.' A blue tooltip at the bottom right says 'Click on a word to get alternative formulations.' In the bottom left corner of the translation area, there is a 'Translate document' button. In the bottom right corner, there are icons for copying, sharing, and downloading.

DeepL | Translator | Linguee

Login

Translate from **English** (detected) ▼

Translate into **German** ▼

Welcome to the webpages of the Data and Web Science Group. We conduct research and offer teaching in the areas data analytics, artificial intelligence, natural language processing, and data integration. The group consists of 7 professors and around 35 researchers and supporting staff members.

Willkommen auf den Webseiten der Data and Web Science Group. Wir forschen und lehren in den Bereichen Datenanalyse, Künstliche Intelligenz, Natürliche Sprachverarbeitung und Datenintegration. Die Gruppe besteht aus 7 Professoren und rund 35 Forschern und unterstützenden Mitarbeitern.

Translate document

Click on a word to get alternative formulations.

Text Summarization

Input Article

Marseille, France (CNN) The French prosecutor leading an investigation into the crash of Germanwings Flight 9525 insisted Wednesday that he was not aware of any video footage from on board the plane. Marseille prosecutor Brice Robin told CNN that "so far no videos were used in the crash investigation." He added, "A person who has such a video needs to immediately give it to the investigators." Robin's comments follow claims by two magazines, German daily Bild and French Paris Match, of a cell phone video showing the harrowing final seconds from on board Germanwings Flight 9525 as it crashed into the French Alps. All 150 on board were killed. Paris Match and Bild reported that the video was recovered from a phone at the wreckage site. ...

Text Summarization Models

Abstractive summarization

Extractive summarization

Generated summary

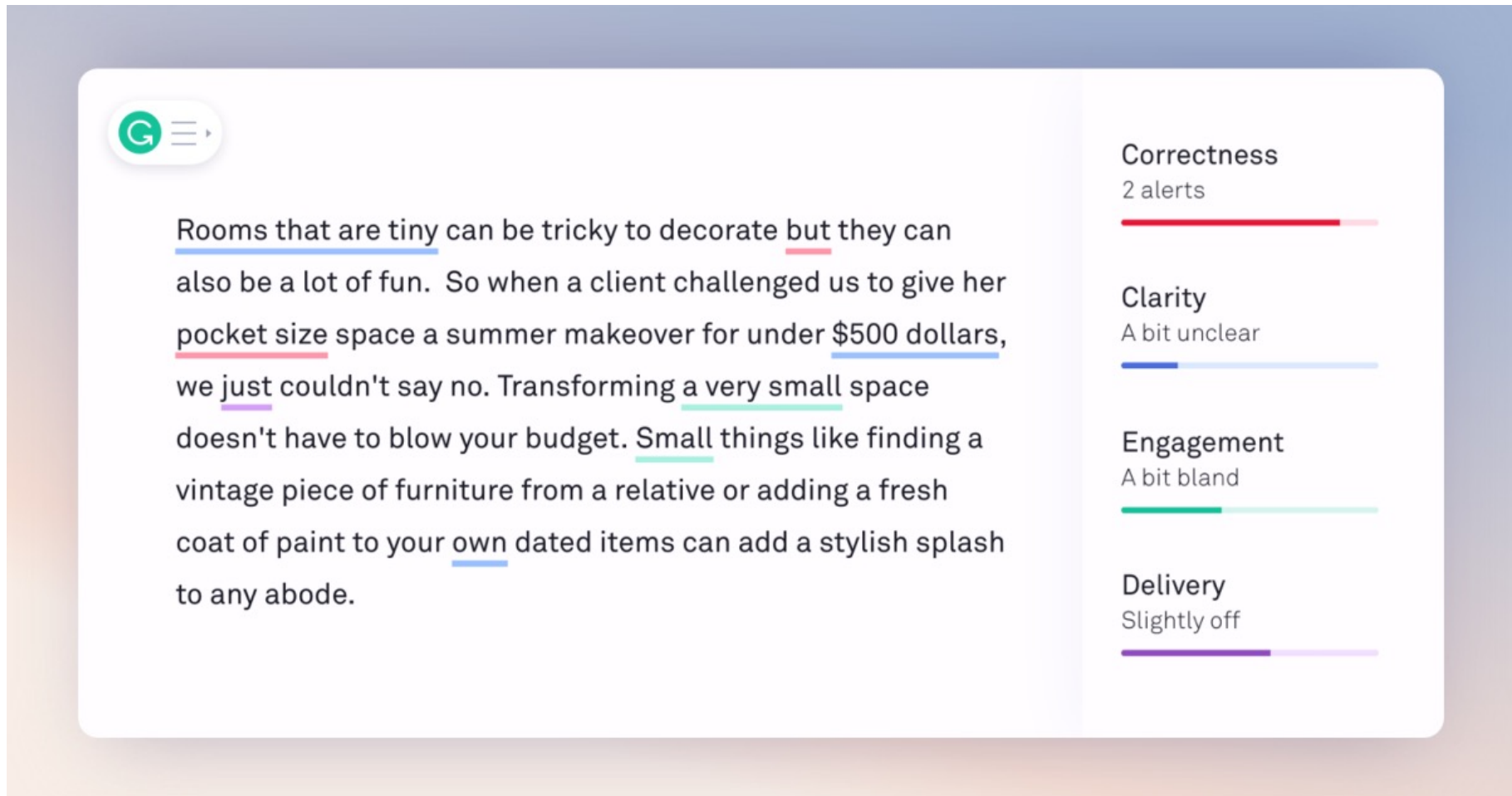
Prosecutor : " So far no videos were used in the crash investigation "

Extractive summary

marseille prosecutor brice robin told cnn that " so far no videos were used in the crash investigation . " robin \s comments follow claims by two magazines , german daily bild and french paris match , of a cell phone video showing the harrowing final seconds from on board germanwings flight 9525 as it crashed into the french alps . paris match and bild reported that the video was recovered from a phone at the wreckage site .

[Image source](#)

Assisted Writing



The screenshot displays a writing assistant interface. On the left, a text area contains a paragraph about decorating small rooms. The text is annotated with colored underlines: blue for 'Rooms that are tiny', red for 'but they can', red for 'pocket size space', blue for '\$500 dollars', purple for 'we just couldn't say no.', green for 'a very small space', cyan for 'Small things', and blue for 'own'. A green circular icon with a 'G' and a menu symbol is in the top left of the text area. On the right, a sidebar lists four metrics, each with a descriptive label and a horizontal progress bar:

- Correctness**: 2 alerts. The progress bar is red and is approximately 80% full.
- Clarity**: A bit unclear. The progress bar is blue and is approximately 20% full.
- Engagement**: A bit bland. The progress bar is green and is approximately 30% full.
- Delivery**: Slightly off. The progress bar is purple and is approximately 40% full.

[Image source](#)

Natural Language Processing Challenges

- Processing natural language is difficult because it requires handling of different aspects of language and communication:
 1. **Phonology**: relations between words and sounds realizing them
 2. **Morphology**: shape and behaviour of words in context
 3. **Syntax**: ordering and grouping of words
 4. **Semantics**: meaning of words (lexical semantics) and larger units of text (compositional semantics)
 5. **Pragmatics**: language use to accomplish goals and intentions
 6. **Discourse**: how text is structured into larger units (paragraphs, documents)
- Understanding text is supported by our **world knowledge**

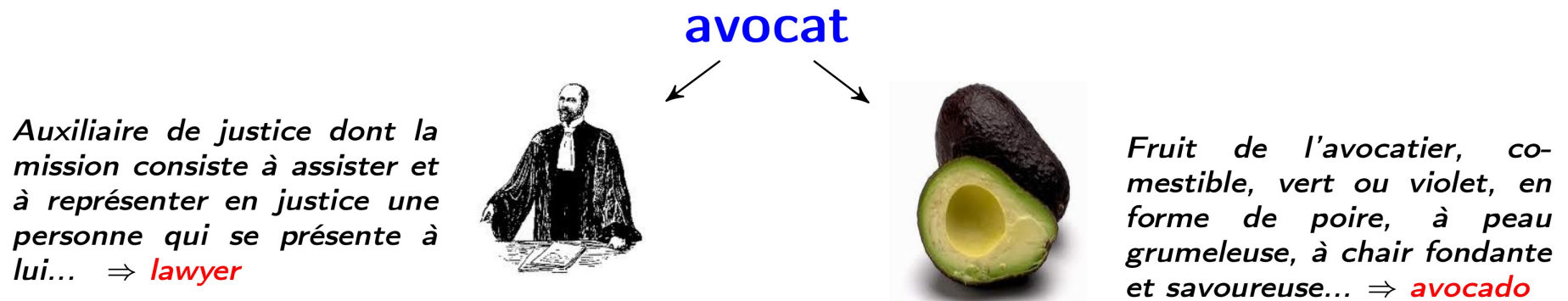
The name of the game is
making sense of ambiguity

... Let's look at some examples!

Language ambiguity

- Ambiguity increases the number of possible interpretations (combinatorial explosion of search space)
- **Categorical ambiguity:**
 - *Flying planes can be dangerous*
- **Word sense ambiguity:**
 - *I saw her run to the bank*
 - *The thief was charged by the police and had to pay a fine.*
- **Structural ambiguity:**
 - *I saw a boy on the hill with a telescope*
- **Referential ambiguity:**
 - *Ann and Lisa gave John and Mark some apples because they liked them*
 - *Lisa gave Ann a present and she said thanks*

Example: (Lexical) Ambiguity in MT



L'**avocat**, riche en lipides, apporte très majoritairement des acides gras insaturés bénéfiques pour la santé cardio-vasculaire.

Natural Language Processing and statistics

- Embrace ambiguity and use statistical methods to appropriately model ambiguity *in context*
- That is, we rely on *probabilistic models built from language data*
- Example: machine translation
 - $P(\text{"maison"} \rightarrow \text{"house"})$ **high**
 - $P(\text{"L'avocat général"} \rightarrow \text{"the general avocado"})$ **low**
- Example: language modeling
 - What word comes after "Every Thursday there is a Schneckenhof ..."?

Language Modeling (1)

- Predict the next word:
 - “Every Thursday there is a Schneckenhof ...”
- Which word is more likely to follow?
 - Meeting?
 - Party?
 - Notebook?
- This suggests:
 - **Some words are more likely to appear than others given some context**
- Probabilistically, a language model (LM) computes the following:
 - $p(\text{meeting} | \text{“Every Thursday there is a Schneckenhof”})$
- **Generally: $p(w_{n+1} | w_1, w_2, \dots, w_n)$**
 - Conditional probability of w_{n+1} given joint distribution of $w_1, w_2, \dots, w_n$

Language Modeling (2)

- In many NLP applications, goal is producing word/token sequences
 - Machine translation
 - Summarization
 - Dialogue systems
- The following task is therefore useful:
 - Given vocabulary $V = \{aardvark, abacus, \dots, zither\}$, **predict probability of sequence of words** $p(w_1, w_2, \dots, w_M)$, with $w_m \in V$
- For example, in machine translation
 - **Input:** *El café negro me gusta mucho.*
- Say a translation system provides the following possible answers
 - *The coffee black pleases me much.* (word-for-word translation)
 - *I love dark coffee.*
- A good language model should say:
 - $P(\textit{The coffee black pleases me much}) < P(\textit{I love dark coffee})$

NLP tasks as optimization problems

- We can view many problems in NLP as an optimization problem

$$\hat{y} = \arg \max_y f(x; y; \theta)$$

- x : input, y : output
 - f : scoring function, i.e. the model
 - $\hat{y}$: the predicted (“best”) output
 - θ : model’s parameter
-
- **Learning**: finding (our estimate of) the best parameters
 - **Inference**: finding (our estimate of) the best output

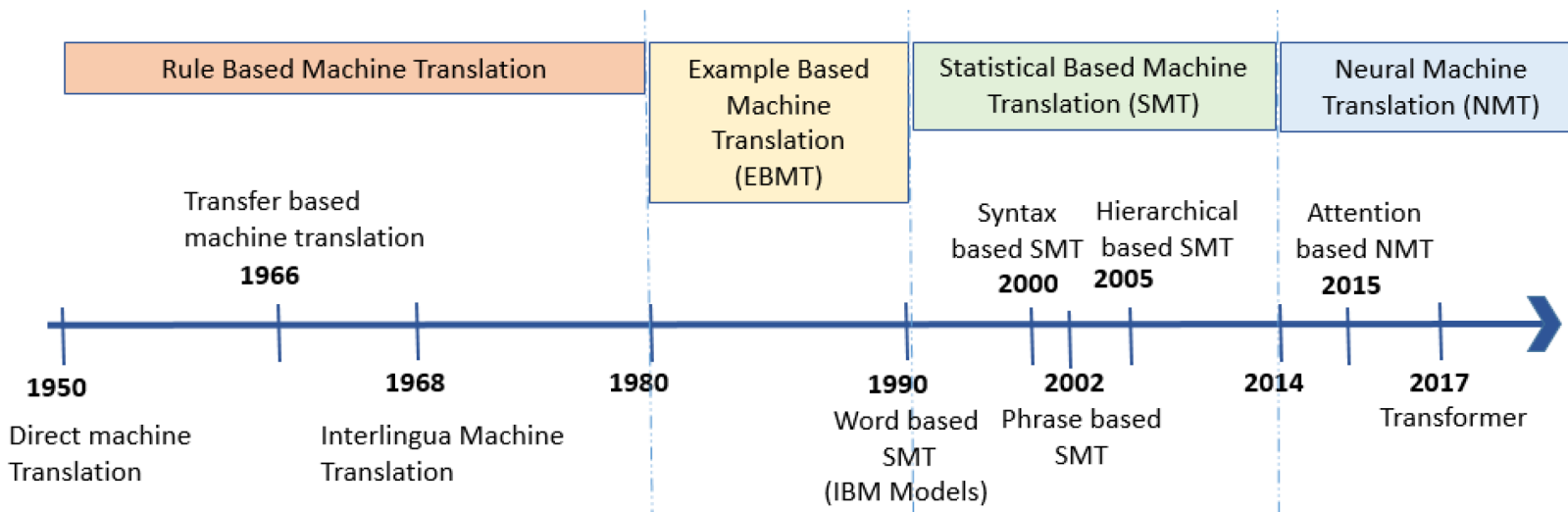
THE CLASSICAL DREAM LINGUISTICS MEET COMPUTATION

The journey ahead

- Let's focus on Machine Translation!
- MT has a long, complex history – I will simplify it and focus on three main different ages / periods
- “Classical” MT => rule-based
- Statistical MT => the data-drive turn
- Neural MT => the current paradigm

A very short history of MT

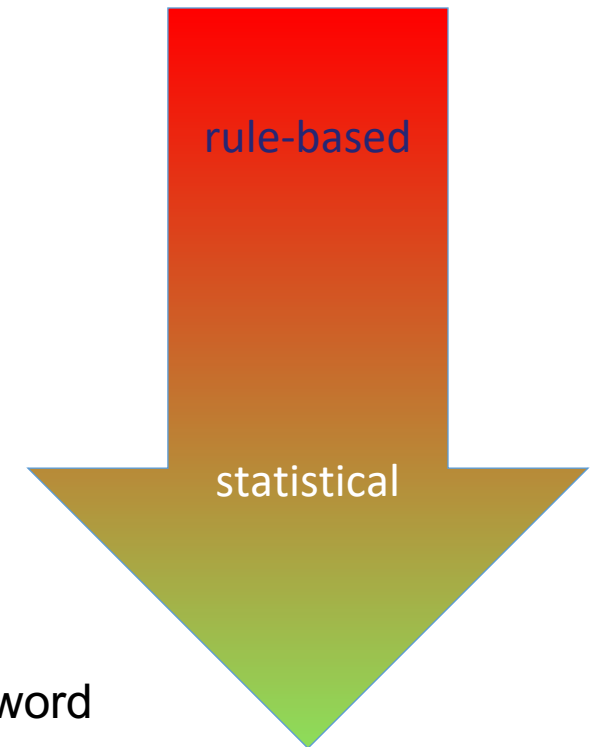
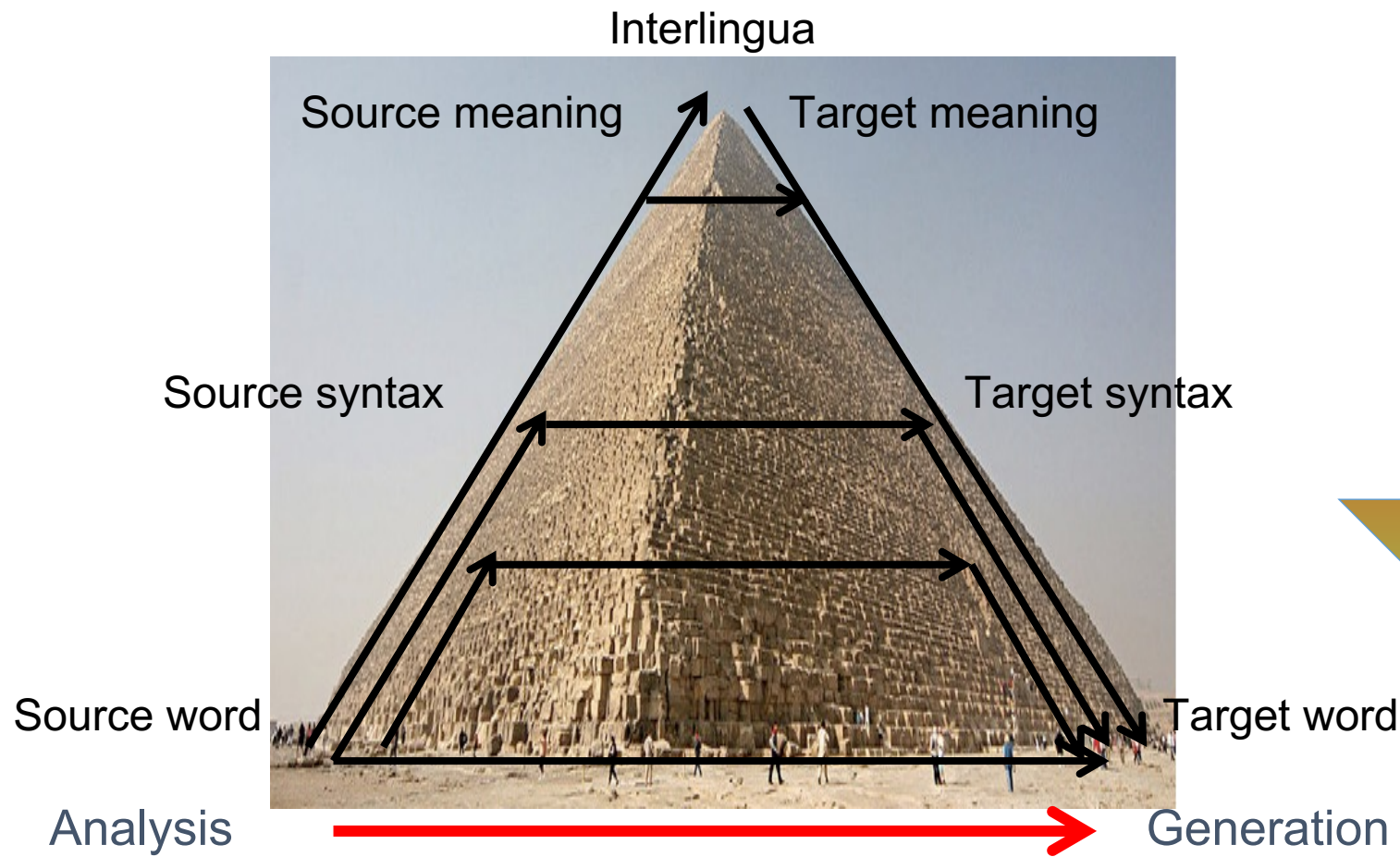
- **1947** – Warren Weaver mentions the possibility of automatic translation in a memo to Norbert Weiner
- **Late 60s – 70s** – First commercially successful MT systems
- **80s – 2000** – Statistical MT (IBM) & Example-based MT (Nagao)
- **2000 – mid 2010s** – adaption of SMT to deal with syntax
- **Mid 2010s** – neural networks / deep learning – neural MT



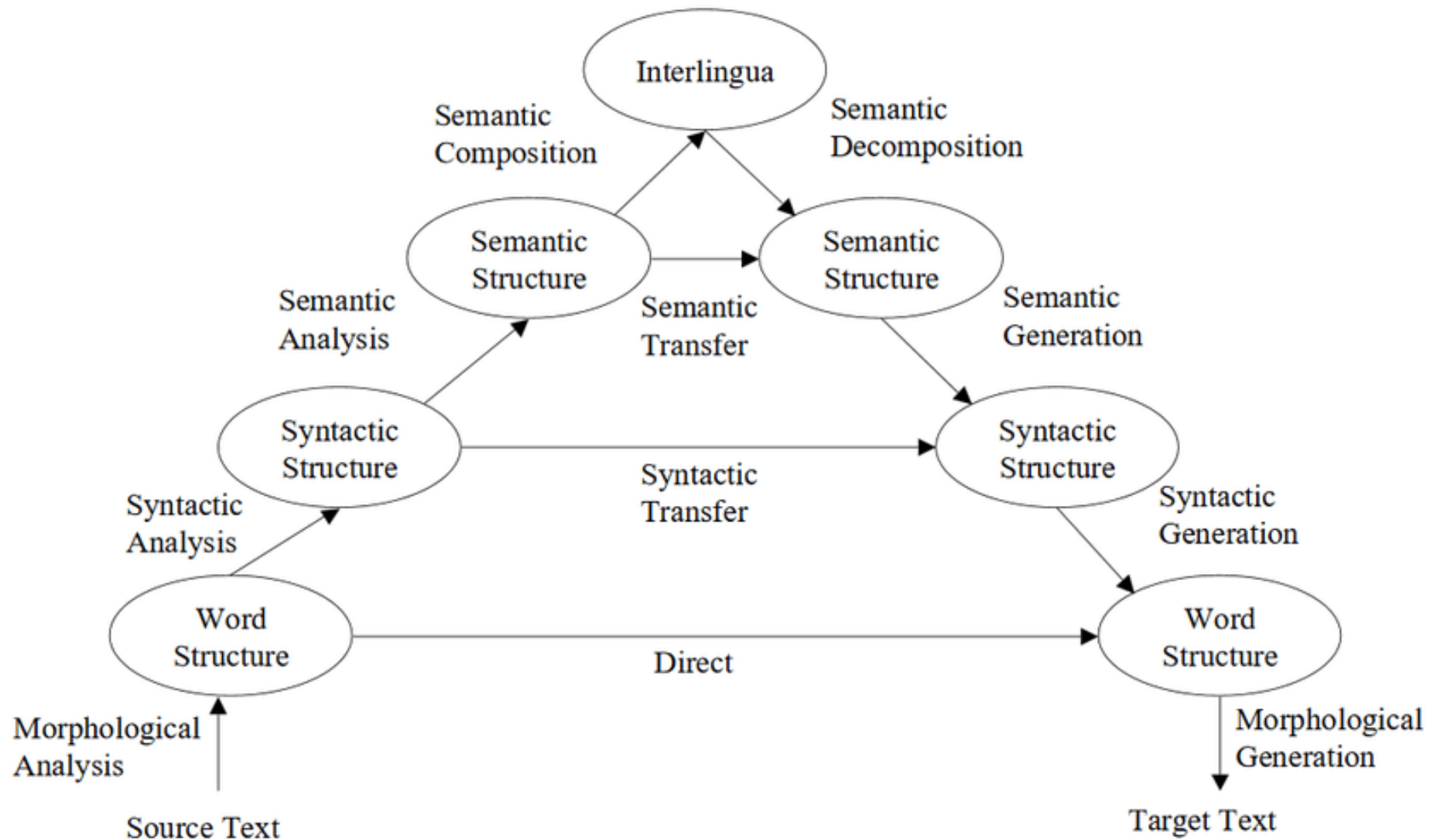
Rule-based MT

- Hand-written transfer rules
- Rules can be based on lexical or structural transfer

The Machine Translation Pyramid



The Vauquois Triangle



Classical approaches

Traditional approach before the SMT revolution:

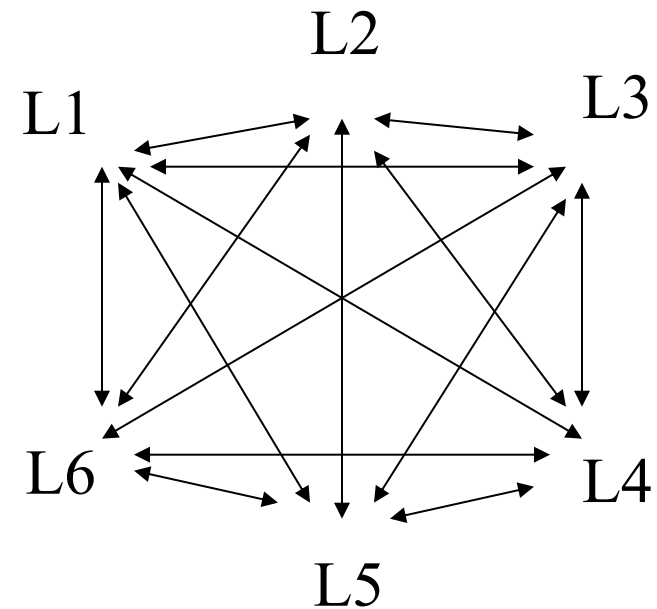
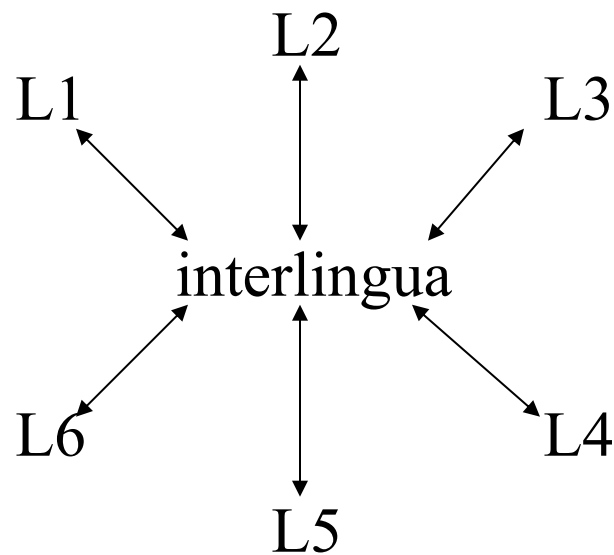
- Transfer- / Syntax-based translation
- Interlingua-based translation

Transfer vs. Interlingua

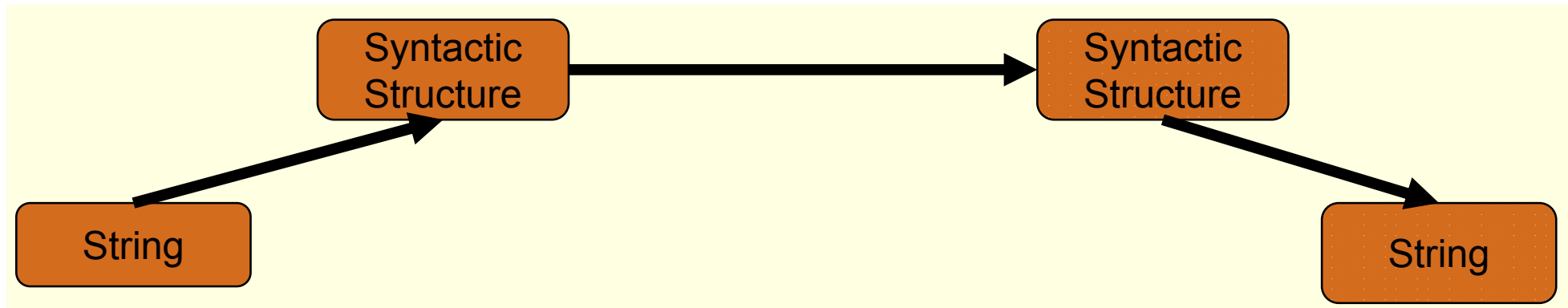
- **Transfer:** Statements in one theory (source language) are mapped into statements in another theory (target language).
- **Interlingua:** Meanings are language independent and can be encoded. They are extracted from source language sentences and rendered as target language sentences.

Interlingua vs. transfer

- Interlingua: N parsers/generators
- Transfer: N^2 transfer systems



Syntactic Transfer



- Source-language syntactic analysis: construct SL analysis tree
- Transfer: Tree-to-tree transformations applied recursively to SL tree: construct TL tree
- Morphological generation
- Consolidation: Applying TL grammar constraints to the TL structure to enforce grammaticality (and fill in underspecified values)

Syntactic Transfer

Example

English grammar

NP → Det N1

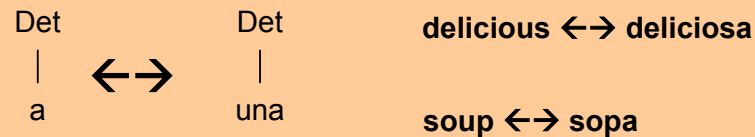
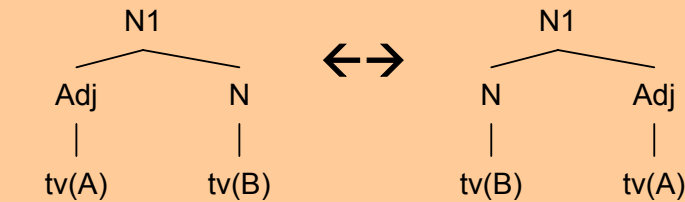
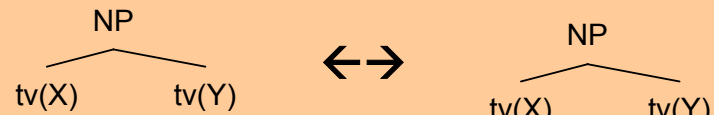
N1 → Adj N

Det → a

Adj → delicious

N → soup

Transfer grammar



Spanish grammar

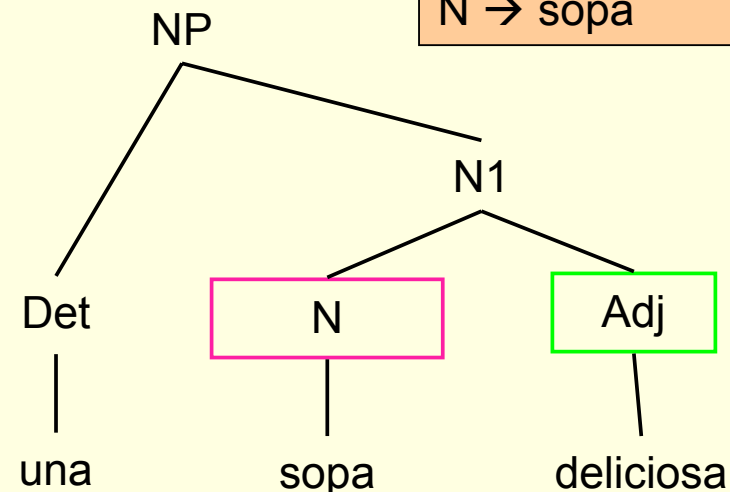
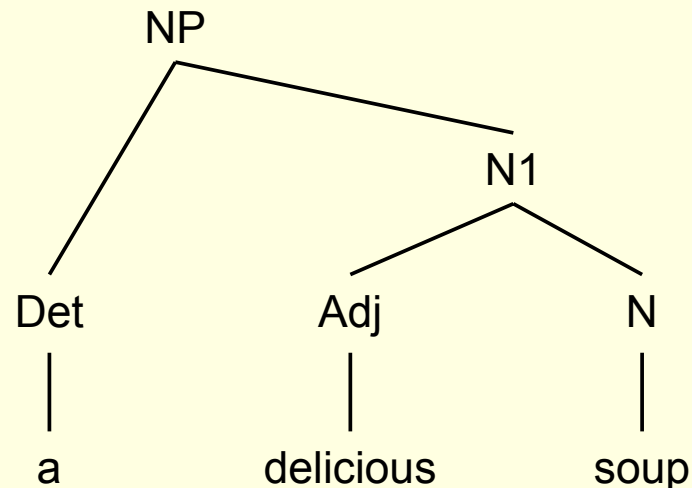
NP → Det N1

N1 → N Adj

Det → una

Adj → deliciosa

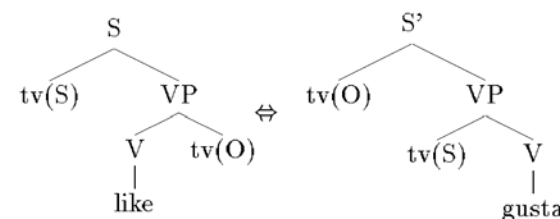
N → sopa



Divergences in syntactic transfer

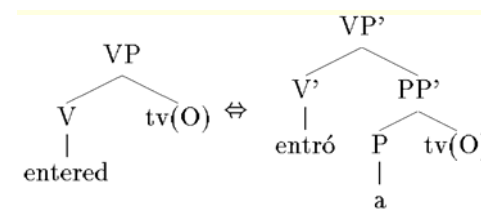
- Thematic divergence

- En: You like her
- Sp: Ella te gusta



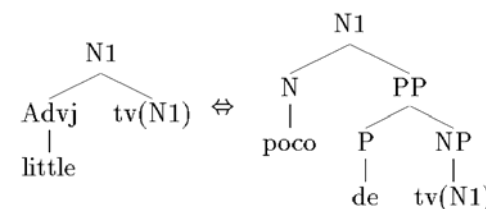
- Structural

- En: Luisa entered the house
- Sp: Luisa entró a la casa



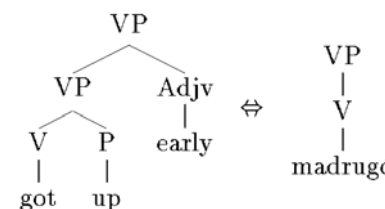
- Categorical

- En: a little bread
- Sp: un poco de pan



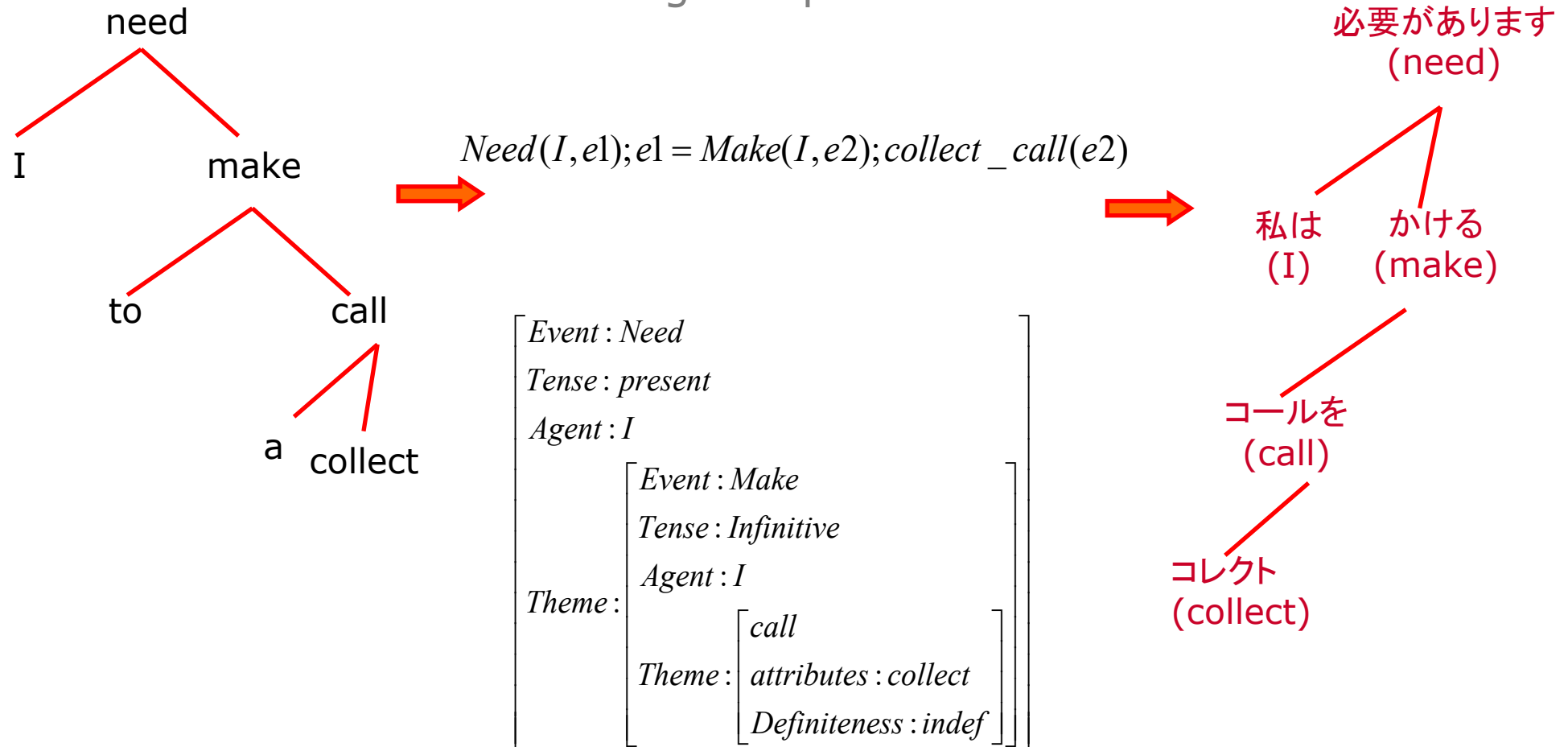
- Lexical gaps (conflational divergence)

- En: Camillo got up early
- Sp: Camillo madrugó



Interlingua

Interlingua representation



Source: Princeton COS401

Rule-based vs. Statistical MT

Rule-based MT

- Hand-written transfer rules
- Rules can be based on lexical or structural transfer
- Pro: firm grip on complex translation phenomena
- Con: Often very labor-intensive, lack of robustness

Statistical MT

- Mainly word or phrase-based translations
- Translation are learned from actual data
- Pro: Translations are learned automatically
- Con: Difficult to model complex translation phenomena

THE DATA TURN SCALE OVER STRUCTURE

Statistical Machine Translation

– Using statistical models

- Create many alternatives, called *hypotheses*
- Give a score to each hypothesis
- Select the best hypothesis to provide the best translation

– Pros

- Easy to build, and it requires no human knowledge
- Can work also with unexpected input
- Good performance when a large amount of training data is available.

– Cons

- Problems handling complex linguistic structures
- Needs parallel data to train the models
- The decisions made by system can be difficult to understand

Statistical MT in a nutshell

- Parallel corpus
- Sentence alignment
- Word alignment
- Statistical model estimation
- Application of the statistical model to unseen text

Parallel Corpus: Training resource for MT

Most historically popular:

- EuroParl: European parliament protocols in 11 languages
- Hansards: Canadian Parliament protocols in French and English

Current ones:

- Meta collections
- Parallel Subtitles
- Parallel webpages



Parallel Corpus: example

English	German
Diverging opinions about planned tax reform	Unterschiedliche Meinungen zur geplanten Steuerreform
The discussion around the envisaged major tax reform continues .	Die Diskussion um die vorgesehene grosse Steuerreform dauert an .
The FDP economics expert , Graf Lambsdorff , today came out in favor of advancing the enactment of significant parts of the overhaul , currently planned for 1999 .	Der FDP - Wirtschaftsexperte Graf Lambsdorff sprach sich heute dafuer aus , wesentliche Teile der fuer 1999 geplanten Reform vorzuziehen .

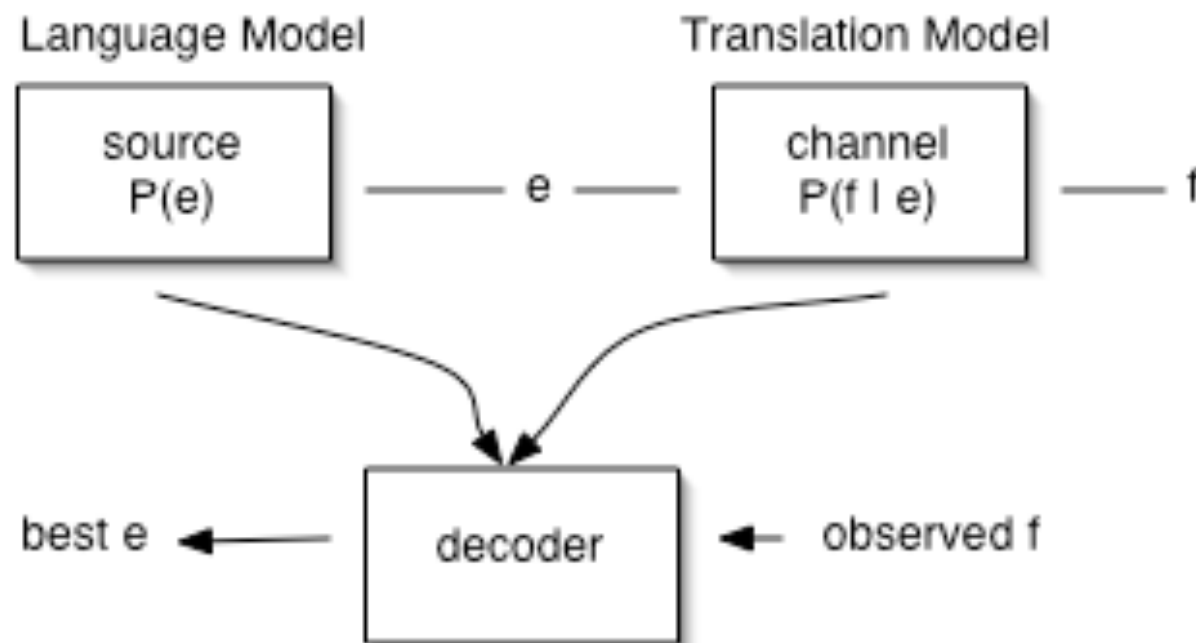
The noisy channel model

- Warren Weaver (1949): “I have a text in front of me which is written in Russian but I am going to pretend that it is really written in English and that it has been coded in some strange symbols. All I need to do is strip off the code in order to retrieve the information contained in the text.”

The noisy channel model for SMT

- To find the best translation we need to compute:

$$\operatorname{argmax}_e P(e|f) = \operatorname{argmax}_e \frac{P(f|e)P(e)}{P(f)} = \operatorname{argmax}_e P(f|e)P(e)$$



Source: Jurafsky & Martin (2009)

Three Elements of a Statistical MT model

– Language model

- Given a target language string e , assigns $P(e)$
- good target language string $\rightarrow$ high $P(e)$
- random word sequence $\rightarrow$ low $P(e)$

– Translation model

- Given a pair of strings $\langle f, e \rangle$, assigns $P(f | e)$
- $\langle f, e \rangle$ look like translations $\rightarrow$ high $P(f | e)$
- $\langle f, e \rangle$ does not look like translations $\rightarrow$ low $P(f | e)$

– Decoding algorithm

- Given a language model, a translation model, and a new sentence f : find translation e maximizing $P(e) \cdot P(f | e)$

On voit Jon à la télévision

	good English? $P(E)$	good match to French? $P(F E)$
Jon appeared in TV.		✓
Appeared on Jon TV.		
In Jon appeared TV.		✓
Jon is happy today.	✓	
Jon appeared on TV.	✓	✓
TV appeared on Jon.	✓	
TV in Jon appeared.		
Jon was not happy.	✓	

Source: Jason Eisner

Language Model: $P(e)$

- Determine the probability of an English sequence $P(e)$, i.e., for MT, it helps selecting picks the **most fluent translation**
- Can modeled using standard n-gram modeling, e.g., bigrams
- $P(\text{"Jon appeared on TV"}) = P(\text{Jon} | \langle s \rangle)$
 $P(\text{appeared} | \text{Jon})$
 $P(\text{on} | \text{appeared})$
 $P(\text{TV} | \text{on})$
 $P(\langle /s \rangle | \text{TV})$

Translation Modeling: $P(f|e)$

- Word-based translation models that rely essentially on the probability that the foreign word f_j is a translation of the English word e_i
- Maximum likelihood estimation
 - Count the number of times f_j is translated as e_i
 - Divide by the number of times e_i is translated.
- Example: translation of German *Haus* from a parallel corpus

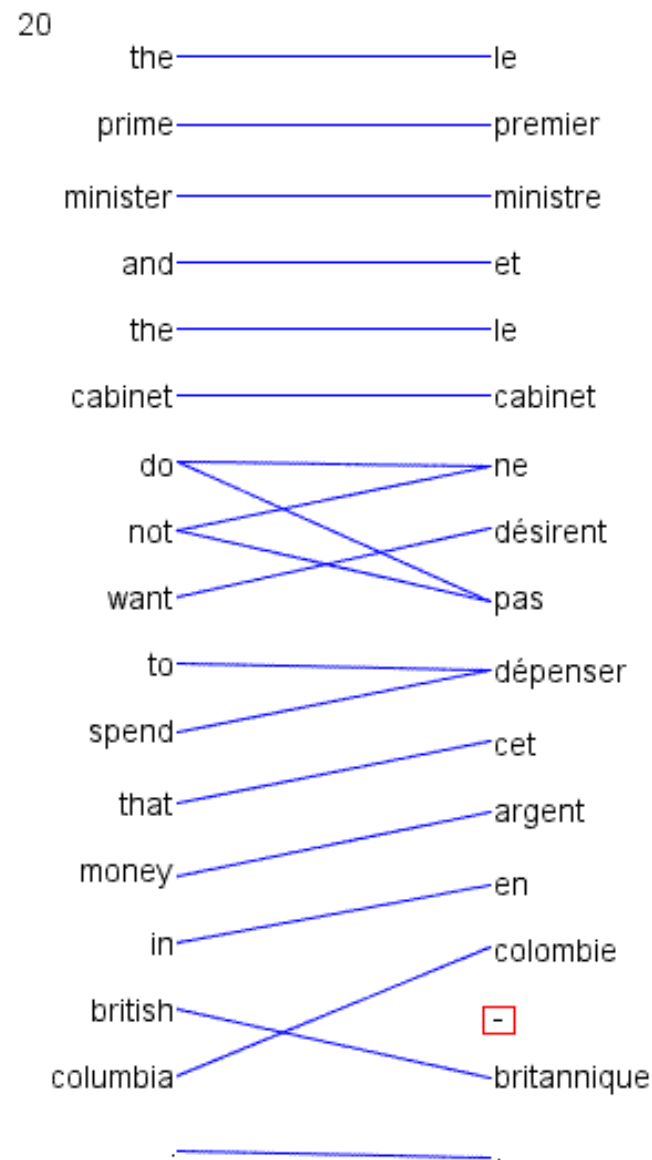
Translation of <i>Haus</i>	Count
house	8,000
building	1,600
home	200
household	150
shell	50

$$p_f(e) = \begin{cases} 0.8 & \text{if } e = \text{house,} \\ 0.16 & \text{if } e = \text{building,} \\ 0.02 & \text{if } e = \text{home,} \\ 0.015 & \text{if } e = \text{household,} \\ 0.005 & \text{if } e = \text{shell.} \end{cases}$$

Source: Koehn SMT book

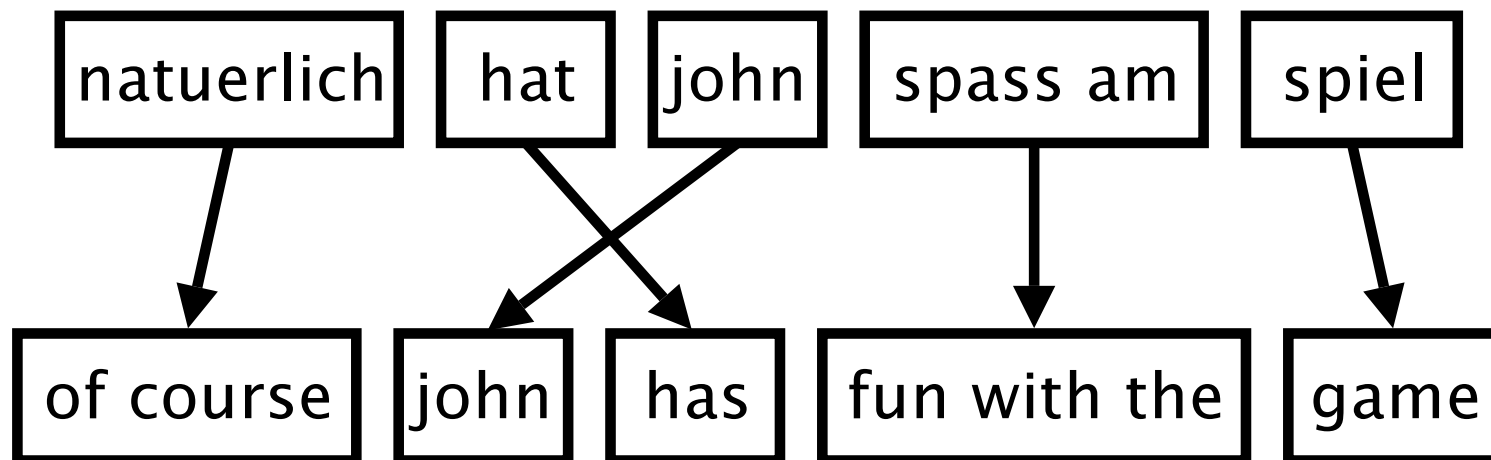
Word alignments

- How to compute $P(f_j | e_i)$ from a parallel corpus? Need to **align** their translations at the word level.
- Generate a word alignment for each sentence pair



Phrase-based SMT

- Foreign input is segmented in phrases
- Each phrase is translated into English
- Phrases are reordered



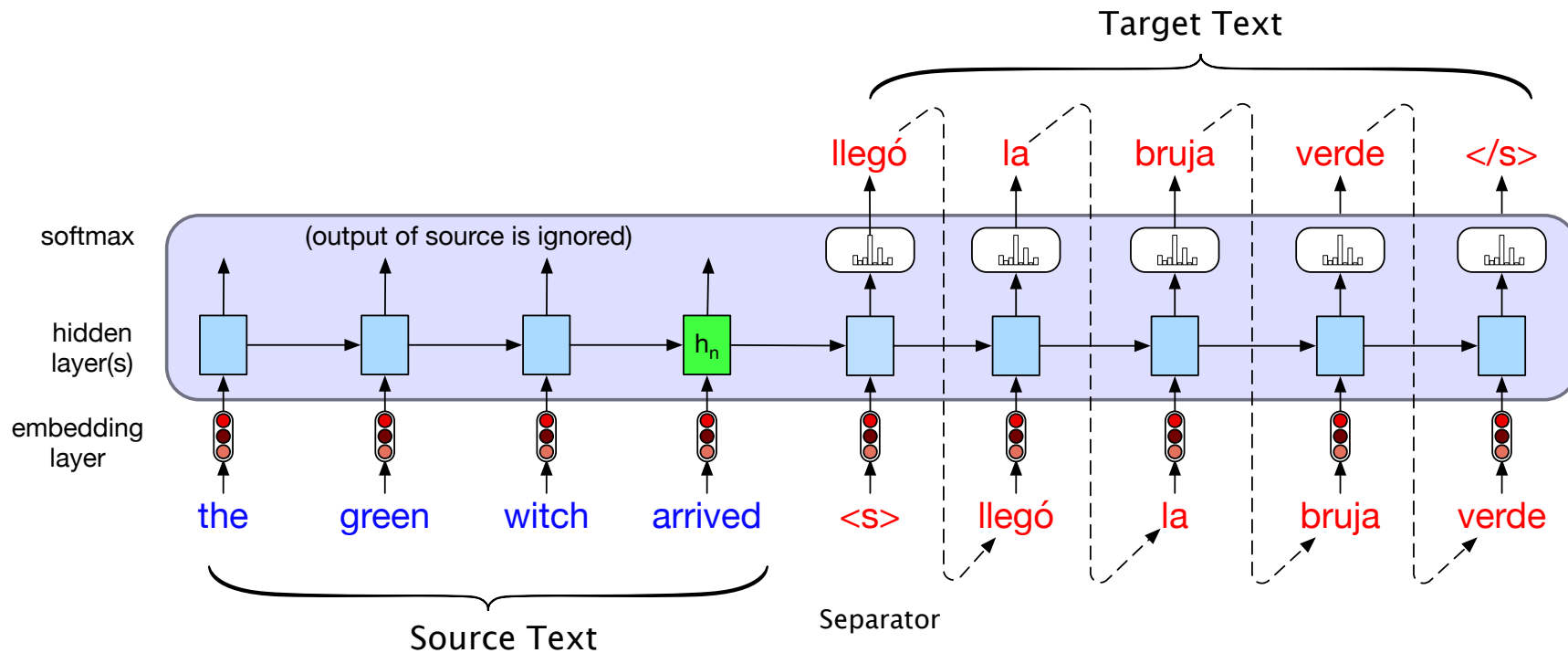
- **NOTE: phrase = “sequence of words”**

Source: Koehn SMT book

Neural Machine Translation

- The problem with SMT: translation and fluency are modeled *separately*
- Ideally, we would like to learn both end-to-end, i.e., directly from translation examples
- Enter deep learning

Encoder-Decoder Model



Source: Martin & Jurafsky

- The encoder reads and summarizes the sentence into a representation, i.e. a vector. The decoder generates the translation based on that “summary”.

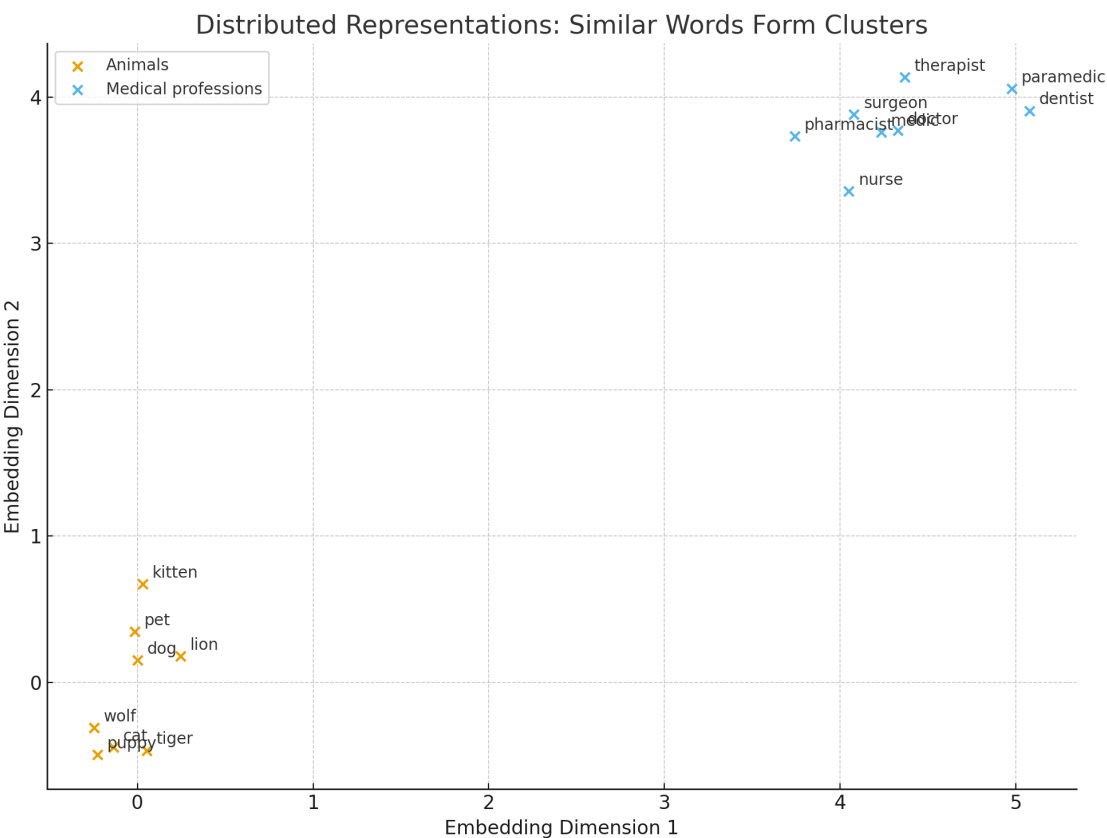
Language representations in neural networks

- Language in neural nets is fundamentally represented in a **distributed** way
- **Distributed meaning:** In neural models, words and sentences aren't stored as symbols but as patterns of activation across many neurons
 - No single neuron “means” dog; meaning emerges from the combination of many.
- **Similarity in vector space:** Words or phrases with similar meanings have similar patterns — the geometry of the vector space encodes linguistic and semantic relationships
 - Example: king – man + woman $\approx$ queen.

Language representations in neural networks

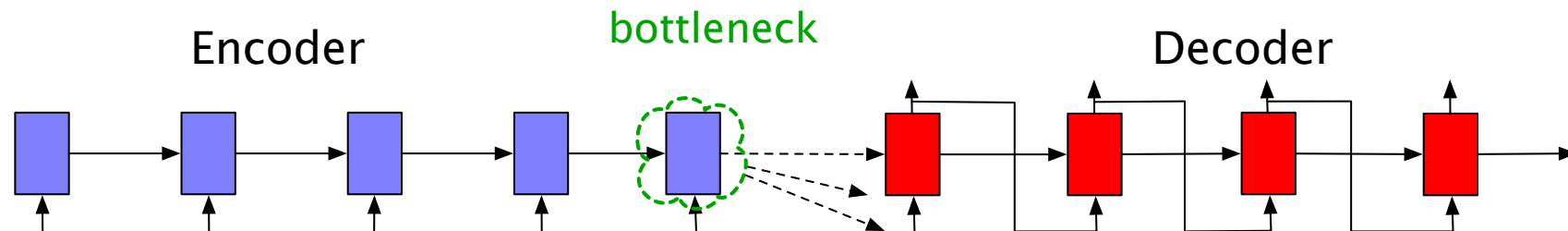
Words Represented as Dense Vectors (Sample of 5 Dimensions)

	Dim 1	Dim 2	Dim 3	Dim 4	Dim 5
dog	-0.161	0.684	-1.786	1.692	0.405
cat	0.381	0.609	0.051	0.606	-0.879
puppy	-1.113	0.215	-0.857	0.302	0.5
lion	-1.44	0.99	-1.412	1.019	1.379
doctor	-0.017	0.649	0.519	-0.219	-0.847
nurse	-1.101	0.102	0.182	-0.431	-0.868
surgeon	-0.23	1.828	2.198		
dentist	1.584	0.323	0.913		



Attention

- Earlier encoder-decoder models had one fixed summary, which was hard for long sentences
- Attention lets the decoder peek at specific words while generating each output
- Instead of memorizing the whole sentence, the model looks back selectively — like glancing at the source text while writing



THE CONFESSION: PERFORMANCE WITHOUT LINGUISTIC STRUCTURES?

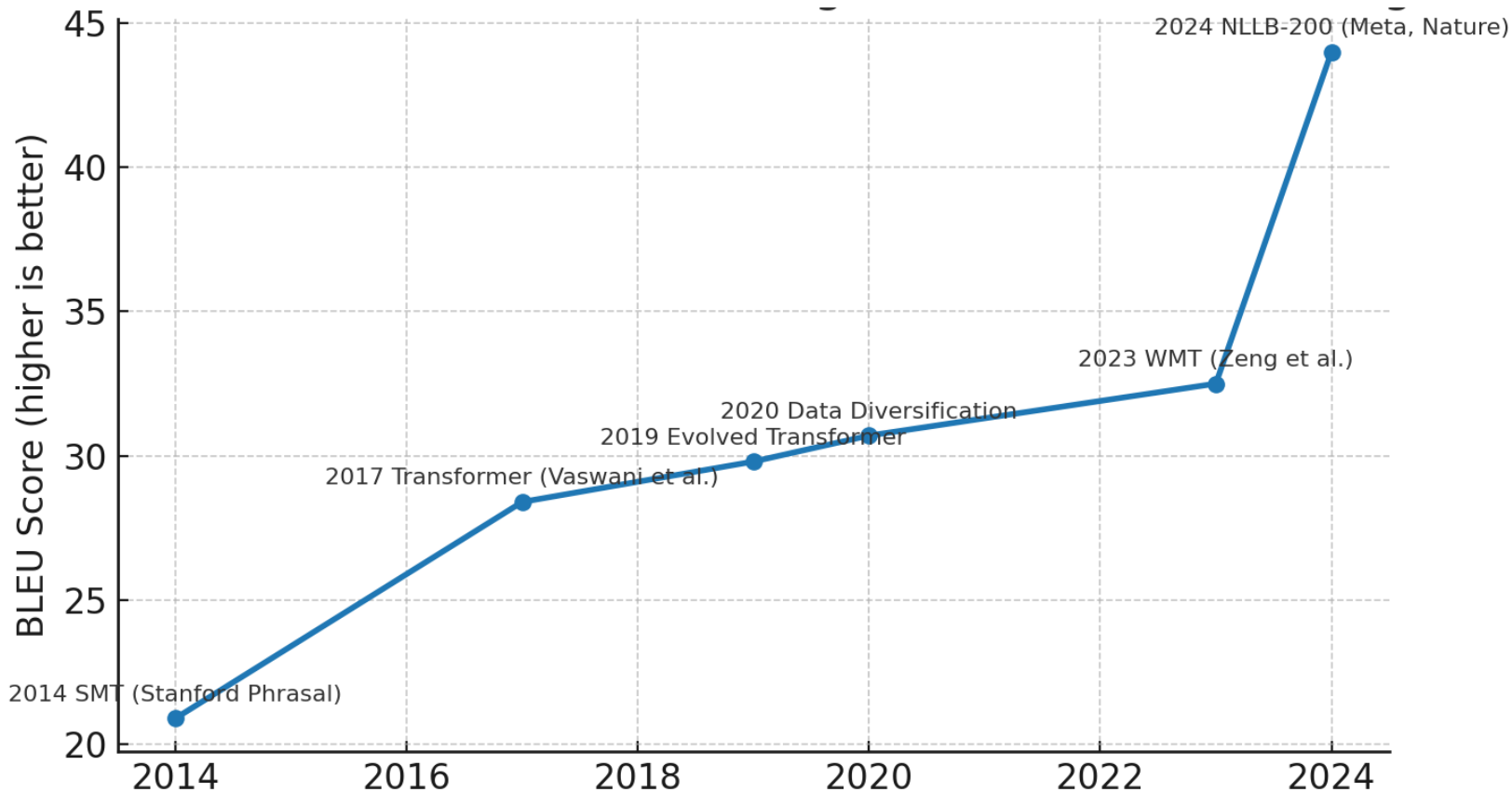
The story so far

Q: how is language represented in the various models?

- We started with explicit linguistic structures, e.g., trees for transfer-based Machine Translation
- We moved to words and phrases with “classical” SMT
- We are left with vectors and matrixes when relying on neural nets

Tradeoff between representation depth and performance

- HOWEVER, despite impoverishing linguistic representations MT performance consistently went up



Nothing new?

- We are still hunted by [Frederick Jelinek](#)'s famous quote

Every time I fire a
linguist, the performance
of the speech recognizer
goes up



Rich Sutton



- Generalizable methods that scale with computation (like search and learning) have out-performed human-designed, domain-specific systems.
- The ‘bitter lesson’ is that leveraging ever-increasing compute trumps embedding our expertise in the system.”
- <http://www.incompleteideas.net/IncIdeas/BitterLesson.html>

Performance without language structures?

- It's perfectly fine to be interested in language as an artifact with specific structures and attributes
- But this doesn't seem to help much as basis for high-performing automated language processing systems
- What is the place of linguistic structures in NLP today?
 - Understanding model representations?
 - Interpretability of the output?
 - Low-resource languages?
 - ...

Performance without understanding?



- Perhaps more troubling is the fact that such performance comes at the cost of not knowing whether/how cutting-edge models exhibit semantic understanding.



On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?

Emily M. Bender*
ebender@uw.edu
University of Washington
Seattle, WA, USA

Angelina McMillan-Major
aymm@uw.edu
University of Washington
Seattle, WA, USA

Timnit Gebru*
timnit@blacknail.org
Black in AI
Palo Alto, CA, USA

Shmargaret Shmitchell
shmargaret.shmitchell@gmail.com
The Aether

ABSTRACT

The past 3 years of work in NLP have been characterized by the development and deployment of ever larger language models, especially for English. BERT, its variants, GPT-2/3, and others, most recently Switch-C, have pushed the boundaries of the possible both through architectural innovations and through sheer size. Using these pretrained models and the methodology of fine-tuning them for specific tasks, researchers have extended the state-of-the-art on a wide array of tasks as measured by leaderboards on specific benchmarks for English. In this paper, we take a step back and ask: How big is too big? What are the possible risks associated with this technology and what paths are available for mitigating those risks? We provide recommendations including weighing the environmental and financial costs first, investing resources into curating and carefully documenting datasets rather than ingesting everything on the web, carrying out pre-development exercises evaluating how the planned approach fits into research and development goals and supports stakeholder values, and encouraging research directions beyond ever larger language models.

CCS CONCEPTS

• Computing methodologies → Natural language processing.

ACM Reference Format:

Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?  In *Conference on Fairness, Accountability, and Transparency (FAT* 21)*, March 3–10, 2021, Virtual Event, Canada. ACM, New York, NY, USA, 14 pages. <https://doi.org/10.1145/3442188.3445922>

1 INTRODUCTION

One of the biggest trends in natural language processing (NLP) has been the increasing size of language models (LMs) as measured by the number of parameters and size of training data. Since 2018

*Joint first authors



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alone, we have seen the emergence of BERT and its variants [39, 70, 74, 113, 146], GPT-2 [106], T-NLG [112], GPT-3 [25], and most recently Switch-C [43], with institutions seemingly competing to produce ever larger LMs. While investigating properties of LMs and how they change with size holds scientific interest, and large LMs have shown improvements on various tasks (§2), we ask whether enough thought has been put into the potential risks associated with developing them and strategies to mitigate these risks.

We first consider environmental risks. Echoing a line of recent work outlining the environmental and financial costs of deep learning systems [129], we encourage the research community to prioritize these impacts. One way this can be done is by reporting costs and evaluating works based on the amount of resources they consume [57]. As we outline in §3, increasing the environmental and financial costs of these models doubly punishes marginalized communities that are least likely to benefit from the progress achieved by large LMs and most likely to be harmed by negative environmental consequences of its resource consumption. At the scale we are discussing (outlined in §2), the first consideration should be the environmental cost.

Just as environmental impact scales with model size, so does the difficulty of understanding what is in the training data. In §4, we discuss how large datasets based on texts from the Internet overrepresent hegemonic viewpoints and encode biases potentially damaging to marginalized populations. In collecting ever larger datasets we risk incurring documentation debt. We recommend mitigating these risks by budgeting for curation and documentation at the start of a project and only creating datasets as large as can be sufficiently documented.

As argued by Bender and Koller [14], it is important to understand the limitations of LMs and put their success in context. This not only helps reduce hype which can mislead the public and researchers themselves regarding the capabilities of these LMs, but might encourage new research directions that do not necessarily depend on having larger LMs. As we discuss in §5, LMs are not performing natural language understanding (NLU), and only have success in tasks that can be approached by manipulating linguistic form [14]. Focusing on state-of-the-art results on leaderboards without encouraging deeper understanding of the mechanism by which they are achieved can cause misleading results as shown

Do Language Models Have Semantics? On the Five Standard Positions

Anders Søgaard
CPAI, University of Copenhagen
Lyngbyvej 2, DK-2100 Copenhagen
soegaard@di.ku.dk

Abstract

We identify five positions on whether large language models (LLMs) and chatbots can be said to exhibit semantic understanding. These positions differ in whether they attribute semantics to LLMs and/or chatbots trained on feedback, what kind of semantics they attribute (inferential or referential), and in *virtue of what* they attribute referential semantics (internal or external causes). This allows for $2^4 = 16$ logically possible positions, only five of which have been argued for. Based on a pairwise comparison of these five positions, we conclude that the better theory of semantics in large language models is, in fact, a sixth combination: Both large language models and chatbots have inferential and referential semantics, grounded in both internal and external causes.

1 Introduction

Large language models (LLMs) are, at heart, trained to solve the so-called *cloze* task, introduced by educational psychologist Wilson L. Taylor in 1953. Each instance of a cloze test is a piece of text in which a word or subword token is omitted:

- (1) Look in thy mirror and tell the face thou —

Taylor would ask his students to guess the missing word and fill in the blank. LLMs are being trained to do the same. Some have argued that because solving cloze tasks is the proper function of LLMs, they cannot possibly acquire *semantic understanding* (Bender and Koller, 2020; Landgrebe and Smith, 2021; Hicks et al., 2024) – by which we shall mean the comprehension and, maybe, production of texts in their linguistic context, but not in their extralinguistic context. Why would anyone think such a capacity is beyond reach for LLMs? Because solving cloze tasks – what LLMs have been trained to do – is essentially a *memorization* task, the kind of thing that parrots can do without

Inferential	Inferential	Ex Referential	Chat Ex Referential
P1	+		
P2	+	+	
P3	+		+
P4	+	+	+
P5	+		

Figure 1: Five positions in the LLM Understanding Debate

semantic understanding (Searle, 1980; Bender and Koller, 2020; Bender et al., 2021).

However, human memory is limited. So is the memory of LLMs. While some of us remember their Shakespeare well enough to infer that the missing word in (1) is *viewer*, others will not, and most LLMs can only memorize a small fraction of the training data (La et al., 2024). In other words, while the proper function of LLMs is to solve cloze tasks, they do not have enough memory to do so. That means they have to adopt another strategy. They have to find a way to guess what is omitted in Shakespearean sonnets or physics textbooks. Such guessing requires semantics.¹

In order for our intuitions to not get in our way, we should remind ourselves that committing to the view that LLMs exhibit semantic understanding does not mean you have to commit to the idea of LLMs teaching themselves semantics. The random initialization of an LLM opens up a space that is astronomically big. All you have to commit to is the idea that the space is big enough for there to be semantic understanding in it, as well as to the idea that an LLM will benefit from accidentally stumbling upon such a capacity.

There are, as far as we can see, five standard

¹Having semantics typically mean outputting words with semantic content, having internal states with semantic content, or understanding words with semantic content. I assume that generation and production turn on the same semantic capacities and will not bake in any assumptions about internal states.

25910

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[On the Dangers of Stochastic Parrots:
Can Language Models Be Too Big?](#)

[Do Language Models Have Semantics?
On the Five Standard Positions](#) (optional
extra material, more recent paper)

That's it folks@

Thanks!

We covered

- A hitchhiker's guide to NLP
- A brief history of MT
- What it tells about the tension we currently experience between the role of linguistic structures and the performance exhibited by state-of the art NLP systems
- Papers / more infos @ <https://www.uni-mannheim.de/dws>

